

Hamza Arfin

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ABOUT

Full-stack developer skilled in Python, Flask, MySQL, REST APIs, database design, and data-driven applications across NLP and recommendation systems.

SKILLS

languages: Python, C, C++, JavaScript, SQL

web: Flask, HTML, CSS, REST APIs, Backend Development, Jekyll

ml: NumPy, Pandas, Scikit-learn, BERT, Transformers, PyTorch

database: MySQL, SQLAlchemy

tools: Git, GitHub

concepts: Data Structures, Algorithms, CRUD, API Integration

EXPERIENCE

FLAME NLP Research Lab, IIIT Delhi

July 2025 - September 2025

Undergraduate Research Intern

New Delhi, India

- Developed and deployed the official lab website using Jekyll
- Built a scalable static website with modular components for managing research updates, publications, and team information
- Structured content using templates and data files to enable easy updates without modifying core code
- Improved usability, performance, and maintainability of the lab's digital presence

PROJECTS

BERT Sarcasm Detection Classifier | [GitHub](#)

May 2026

- Fine-tuned bert-base-uncased on News Headlines data with stratified split and weighted loss
- Optimized threshold to 0.3965, reaching 93.77% F1, 94.06% accuracy, and 0.983 ROC-AUC
- Built FastAPI /predict endpoint and UI returning confidence, explanations, and tags
- Generated reproducible evaluation artifacts including confusion matrix, ROC/PR curves, threshold-F1 curve, and training logs

Hybrid Movie Recommendation System | [GitHub](#)

March 2026

- Built a recommendation platform using Python, Flask, HTML, CSS, JavaScript
- Implemented hybrid recommender using KNN collaborative filtering on the MovieLens dataset (100k+ ratings)
- Integrated TMDB API and built REST endpoints serving movie metadata and posters

Streamly Streaming Aggregator | [GitHub](#)

November 2025

- Developed streaming search platform using Flask and TMDB API
- Implemented real-time search suggestions and optimized API requests with rate limiting

PythonMySQL Hospital Database System | [GitHub](#)

October 2023

- Built command-line hospital record system using Python and MySQL with automatic database and table setup
- Implemented CRUD workflows for doctors, nurses, workers, and patients, including patient admission and discharge handling
- Used parameterized SQL queries, input validation, environment-based credentials, and tabular console output for safer operations

EDUCATION

Jamia Hamdard University

2023 - 2027

Bachelor of Technology in Computer Science and Engineering

New Delhi, India

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering

Dev Samaj Modern School - 2

2021 - 2023

Senior Secondary School

New Delhi, India

Dev Samaj Modern School - 2

2019 - 2021

Secondary School

New Delhi, India